

Key Insights from the Heart Disease Dataset

1. Dataset Overview

- The dataset contains **918 patient records** with **12 features** related to demographic, clinical, and ECG measurements.
- The target variable is **HeartDisease**, indicating whether a patient has heart disease.

2. Gender Distribution

- The dataset is heavily **male-dominated**:
 - **Male:** 725 (~79%)
 - **Female:** 193 (~21%)
- This indicates that the dataset is imbalanced with respect to gender.

3. Age Distribution

- Most patients fall within the **40–60 years** age group.
- The distribution is approximately bell-shaped, with fewer very young and very old patients.

4. Resting Blood Pressure

- Most individuals have resting blood pressure values in the normal to moderately elevated range.
- A few extreme values indicate the presence of potential outliers.

5. Cholesterol Levels

- Cholesterol values show a wide spread across patients.
- Some observations have extremely low or zero cholesterol values, which may require data validation or preprocessing before model building.

6. Maximum Heart Rate (MaxHR)

- Maximum heart rate varies considerably among patients.
- Older patients generally tend to have lower maximum heart rates.

7. Fasting Blood Sugar

- The majority of patients have **normal fasting blood sugar (FastingBS = 0)**.
- Only a smaller proportion have elevated fasting blood sugar.

8. Heart Disease Distribution

- The target variable is relatively balanced, making it suitable for predictive modeling with minimal class imbalance issues.

9. Correlation Analysis

- **Age** has a **negative correlation** with **Maximum Heart Rate (MaxHR)**.
- **Oldpeak** shows a positive relationship with heart disease.
- **MaxHR** has a negative relationship with heart disease.
- Most numerical features exhibit weak-to-moderate correlations, suggesting that multiple variables collectively influence heart disease.

10. Age vs Maximum Heart Rate

- Joint plots indicate that **maximum heart rate decreases as age increases**, which aligns with known physiological trends.

11. Heart Disease vs Age

- Violin plots suggest that patients diagnosed with heart disease are generally older than those without the disease.

12. Heart Disease vs Cholesterol

- Cholesterol alone does not clearly separate patients with and without heart disease, indicating that it should be considered alongside other clinical variables.
13. **Categorical Feature Distribution**
- Chest pain type, resting ECG, and ST slope exhibit different distributions, suggesting these features may be useful predictors for heart disease classification.
14. **Data Quality Observation**
- Numerical features such as cholesterol and resting blood pressure contain outliers.
 - Proper preprocessing (outlier handling, feature scaling, and missing value checks if applicable) is recommended before machine learning.

Key Insights from the Heart Disease Dataset

- **Total Records: 918**
- **Total Features: 12**
- **Numerical Features: 7**
- **Categorical Features: 5**
- **Missing Values: 0** (918 non-null values in every column)
- **Duplicate Records: 0** (if duplicates were removed during preprocessing)

Dataset Composition

- **918 patients** were included in the analysis.
- All **12 variables** contain complete information, resulting in **100% data completeness**.
- Dataset memory usage is approximately **86.2 KB**.

Numerical Feature Summary

Feature	Description
Age	Patient age
RestingBP	Resting blood pressure
Cholesterol	Serum cholesterol
FastingBS	Fasting blood sugar
MaxHR	Maximum heart rate
Oldpeak	ST depression induced by exercise
HeartDisease	Target variable (0 = No, 1 = Yes)

Correlation Insights

The strongest correlations with **HeartDisease** are:

Feature	Correlation with Heart Disease
Oldpeak	+0.404

Feature	Correlation with Heart Disease
MaxHR	−0.400
Age	+0.282
FastingBS	+0.267
Cholesterol	−0.233
RestingBP	+0.108

Interpretation

- **Oldpeak ($r = 0.404$)** has the **strongest positive relationship** with heart disease. Patients with higher ST depression are more likely to have heart disease.
- **Maximum Heart Rate ($r = -0.400$)** has the **strongest negative relationship** with heart disease. Lower maximum heart rate is associated with a higher probability of heart disease.
- **Age ($r = 0.282$)** shows a moderate positive relationship, indicating that heart disease risk generally increases with age.
- **Fasting Blood Sugar ($r = 0.267$)** also contributes positively, suggesting elevated blood sugar is associated with greater heart disease risk.
- **Cholesterol ($r = -0.233$)** shows only a weak negative correlation, implying cholesterol alone is not a strong predictor in this dataset.
- **Resting Blood Pressure ($r = 0.108$)** has a very weak positive relationship with heart disease.

Relationships Among Numerical Features

- **Age vs MaxHR: -0.382**
 - Older patients generally achieve lower maximum heart rates.
- **Age vs Oldpeak: $+0.259$**
 - ST depression tends to increase with age.
- **Age vs RestingBP: $+0.254$**
 - Resting blood pressure increases slightly with age.
- **Cholesterol vs MaxHR: $+0.236$**
 - Weak positive association.
- **Cholesterol vs FastingBS: -0.261**
 - Weak negative relationship.
- **Oldpeak vs MaxHR: -0.161**
 - Higher ST depression is associated with lower maximum heart rate.

Data Quality Insights

- **918/918 (100%)** records have complete values.
- No missing values were found in any feature.
- The dataset includes both numerical and categorical variables, making it suitable for:
 - Classification
 - Correlation analysis
 - Feature engineering
 - Predictive machine learning